

CO21BTECH11002

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Theory Assignment 1

Q1 (2.1)

To show that the prefix order " $\leq$ " is a partial order on Σ^* , we need to show that " $\leq$ " is reflexive, asymmetric and transitive.

1. Reflexivity

We need to show that for every word $v \in \Sigma^*$, we have $v \leq v$.

By definition, $v \leq w$ means v is a prefix of w .

A word is always a prefix of itself (since it starts with itself). Thus, $v \leq v$ holds for all v , proving reflexivity.

2. Antisymmetry

We need to show that if $v \leq w$ and $w \leq v$, then $v = w$.

- $v \leq w$ means that v is a prefix of w , so w starts with v .
- $w \leq v$ means that w is a prefix of v , so v starts with w .

If two words are prefixes of each other, they must be the same length (otherwise, the shorter word would not contain the extra characters of the longer one). Since both words are the same length and contain the same characters, we conclude $v = w$ proving antisymmetry.

3. Transitivity

We need to show that if $v \leq w$ and $w \leq u$, then $v \leq u$.

- $v \leq w$ means v is a prefix of w , so w starts with v .
- $w \leq u$ means w is a prefix of u , so u starts with w .

Since w starts with v and u starts with w , it follows that u must start with v as well. Thus, $v \leq u$, proving transitivity.

Since the prefix order satisfies reflexivity, antisymmetry, and transitivity, it is a partial order on Σ^* .

Q2 (3.1)

$$s = r_1(x)r_2(y)w_1(y)r_3(z)w_3(z)r_2(x)w_2(z)w_1(x)$$

Assuming T_0 as the initializer transaction, T_∞ as the final reader transaction.

a) $H[s]$:

$$H_s[w_0(x)] = f_{0x}()$$

$$H_s[w_0(y)] = f_{0y}()$$

$$H_s[w_0(z)] = f_{0z}()$$

$$r_1(x) \text{ depends on } w_0(x). \text{ Thus, } H_s[r_1(x)] = H_s[w_0(x)] = f_{0x}()$$

$$\text{Similarly, } r_2(y) = f_{0y}()$$

Now, $w_1(y)$ depends on all reads by transaction 1, i.e., $r_1(x)$. Thus,

$$H_s[w_1(y)] = f_{1y}(f_{0x}())$$

$$r_3(z) \text{ depends on } w_0(x). \text{ Thus } H_s[r_3(z)] = f_{0z}()$$

$$w_3(z) \text{ depends on } r_3(z). \text{ Thus, } H_s[w_3(z)] = f_{3z}(f_{0z}())$$

$$r_2(x) \text{ depends on } w_0(x). \text{ Thus } H_s[r_2(x)] = f_{0x}()$$

$$w_2(z) \text{ depends on } r_2(y), r_2(x). \text{ Thus, } H_s[w_2(z)] = f_{2z}(f_{0x}(), f_{0y}())$$

$$w_1(x) \text{ depends on } r_1(x). \text{ Thus, } H_s[w_1(x)] = f_{1x}(f_{0x}())$$

Now, $H[s]$ is the herbrand semantics for the last write on each object.

$$H[s](x) = H_s[w_1(x)] = f_{1x}(f_{0x}())$$

$$H[s](y) = H_s[w_1(y)] = f_{1y}(f_{0x}())$$

$$H[s](z) = H_s[w_3(z)] = f_{2z}(f_{0x}(), f_{0y}())$$

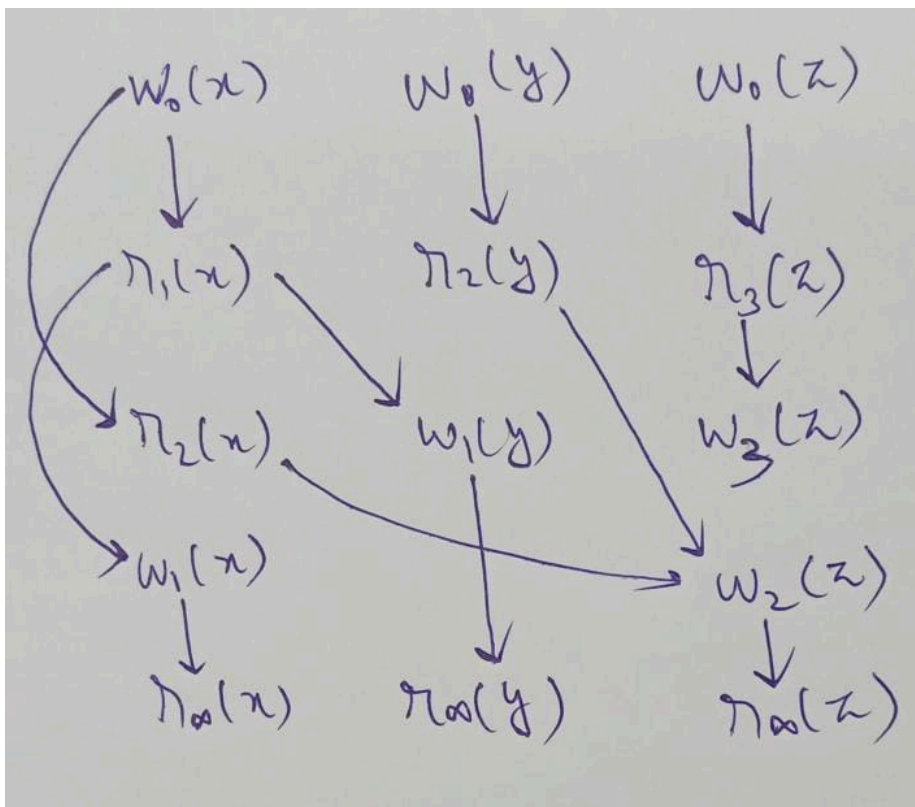
b) RF, LRF

Using the dependence relations from previous part,

$$RF(s) = \{(t0, x, t1), (t0, y, t2), (t0, z, t3), (t0, x, t2), (t1, x, t\infty), (t1, y, t\infty), (t2, z, t\infty)\}$$

$$LRF(s) = \{(t0, x, t1), (t0, y, t2), (t0, x, t2), (t1, x, t\infty), (t1, y, t\infty), (t2, z, t\infty)\}$$

c) Step Graph



Q3 (3.3)

Function corresponding to write step: $f(x) = x + 1$

$$s = r_3(z)r_1(y)w_3(z)w_1(y)r_1(x)r_2(y)w_2(y)w_1(x)r_2(x)w_2(x)c_1c_2c_3$$

$$s' = r_3(z)w_3(z)r_2(y)w_2(y)r_1(y)w_1(y)r_2(x)w_2(x)r_1(x)w_1(x)c_3c_2c_1$$

Assuming that writes of a transaction, say $w_i(obj)$, depends only on the last value of obj read by the transaction i , since $w_i(obj)$ will only increment the value of obj based on its knowledge about obj which can only come from the last read of obj by the same transaction.

Assuming x, y, z are initialized as x_0, y_0, z_0 respectively by the initializer transaction.

$$\text{So, } H_s[w_0(x)] = f_{0x}() = x_0, H_s[w_0(y)] = f_{0y}() = y_0, H_s[w_0(z)] = f_{0z}() = z_0.$$

For s:

$$H_s[r_3(z)] = H_s[w_0(z)] = z_0$$

$$H_s[r_1(y)] = H_s[w_0(y)] = y_0$$

$$H_s[w_3(z)] = f_{3z}(H_s[r_3(z)]) = f(z_0) = z_0 + 1$$

$$H_s[w_1(y)] = f_{1y}(H_s[r_1(y)]) = f(y_0) = y_0 + 1$$

$$H_s[r_1(x)] = H_s[w_0(x)] = x_0$$

$$H_s[r_2(y)] = H_s[w_1(y)] = y_0 + 1$$

$$H_s[w_2(y)] = f_{2y}(H_s[r_2(y)]) = f(y_0 + 1) = y_0 + 2$$

$$H_s[w_1(x)] = f_{1x}(H_s[r_1(x)]) = f(x_0) = x_0 + 1$$

$$H_s[r_2(x)] = H_s[w_1(x)] = x_0 + 1$$

$$H_s[w_2(x)] = f_{2x}(H_s[r_2(x)]) = f(x_0 + 1) = x_0 + 2$$

H[s]:

$$H[s](x) = H_s[w_2(x)] = x_0 + 2$$

$$H[s](y) = H_s[w_2(y)] = y_0 + 2$$

$$H[s](y) = H_s[w_3(z)] = z_0 + 1$$

For s':

$$s' = r_3(z)w_3(z)r_2(y)w_2(y)r_1(y)w_1(y)r_2(x)w_2(x)r_1(x)w_1(x)c_3c_2c_1$$

$$H_{s'}[r_3(z)] = H_{s'}[w_0(z)] = z_0$$

$$H_{s'}[w_3(z)] = f_{3z}(H_{s'}[r_3(z)]) = f(z_0) = z_0 + 1$$

$$H_{s'}[r_2(y)] = H_{s'}[w_0(y)] = y_0$$

$$H_{s'}[w_2(y)] = f_{2y}(H_{s'}[r_2(y)]) = f(y_0) = y_0 + 1$$

$$H_{s'}[r_1(y)] = H_{s'}[w_2(y)] = y_0 + 1$$

$$H_{s'}[w_1(y)] = f_{1y}(H_{s'}[r_1(y)]) = f(y_0 + 1) = y_0 + 2$$

$$H_{s'}[r_2(x)] = H_{s'}[w_0(x)] = x_0$$

$$H_{s'}[w_2(x)] = f_{2x}(H_{s'}[r_2(x)]) = f(x_0) = x_0 + 1$$

$$H_{s'}[r_1(x)] = H_{s'}[w_2(x)] = x_0 + 1$$

$$H_{s'}[w_1(x)] = f_{1x}(H_{s'}[r_1(x)]) = f(x_0 + 1) = x_0 + 2$$

H[s']:

$$H[s'](x) = H_{s'}[w_1(x)] = x_0 + 2$$

$$H[s'](y) = H_{s'}[w_1(y)] = y_0 + 2$$

$$H[s'](y) = H_{s'}[w_3(z)] = z_0 + 1$$

Since $H[s] = H[s']$, the schedules s and s' are final state serializable as per the given function corresponding to the write steps.